



Impact of nirsevimab on the epidemiology of pediatric RSV in Belgium in season 2024-2025

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INTRODUCTION

- RSV is one of the most prevalent causes of respiratory infections in infants globally. **In Belgium, yearly more than 10.000 children under 5 years are hospitalized due to a proven RSV infection**, with almost 7.000 below the age of 1 year ¹.
- A monoclonal antibody, nirsevimab (Beyfortus®), is available and reimbursed in Belgium since October 2024** for all children encountering their first RSV season (at birth + catch-up campaign for children <6 months old at the start of the season)

AIM

- Assess the RSV pediatric **burden of disease and monitor impact** of preventive tools on incidences
- Measure **immunization effectiveness** using test-negative design

METHOD

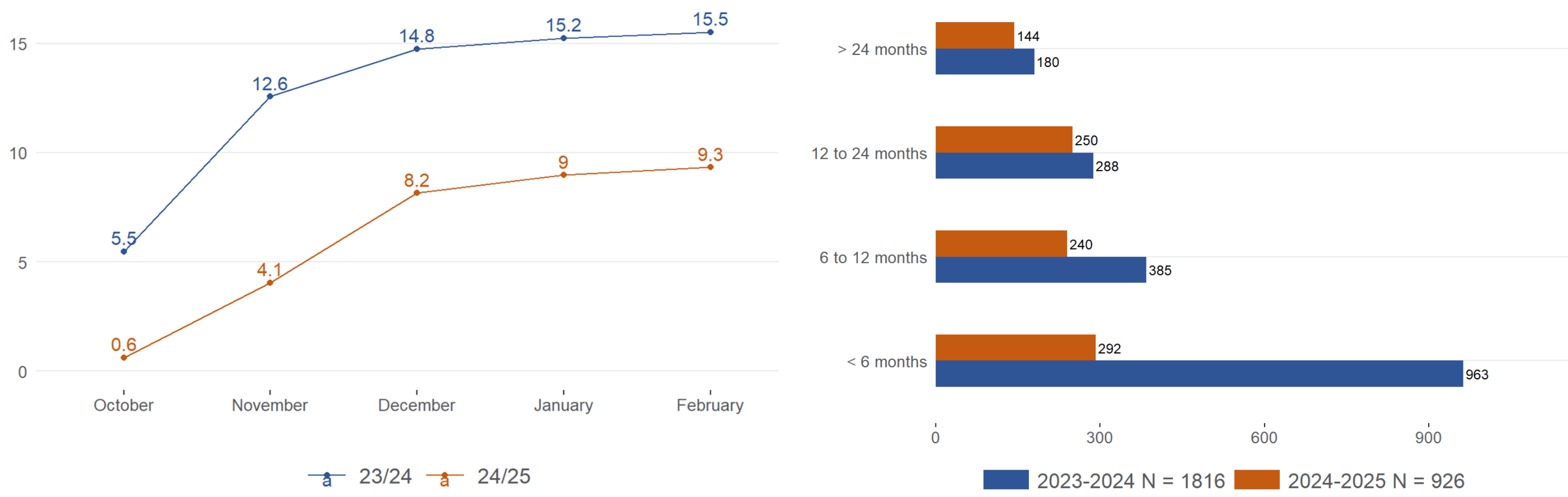
- Broad national surveillance network** (31 hospitals) of pediatric RSV hospitalizations (covering 43% of population, 67% for PICU)
- Adaptation of the **SARI hospital surveillance** (9 hospitals), new case definition and questionnaire
 - RSVPed study** in 25 hospitals*: inclusion of all proven RSV hospital admissions <5 year-old meeting the same case definition and hospitalized Oct-Feb; same questionnaire

Data collected in **pre-prevention season (2023-24)** and the **first year of nirsevimab implementation (2024-25)**

* 3 hospitals participated to both

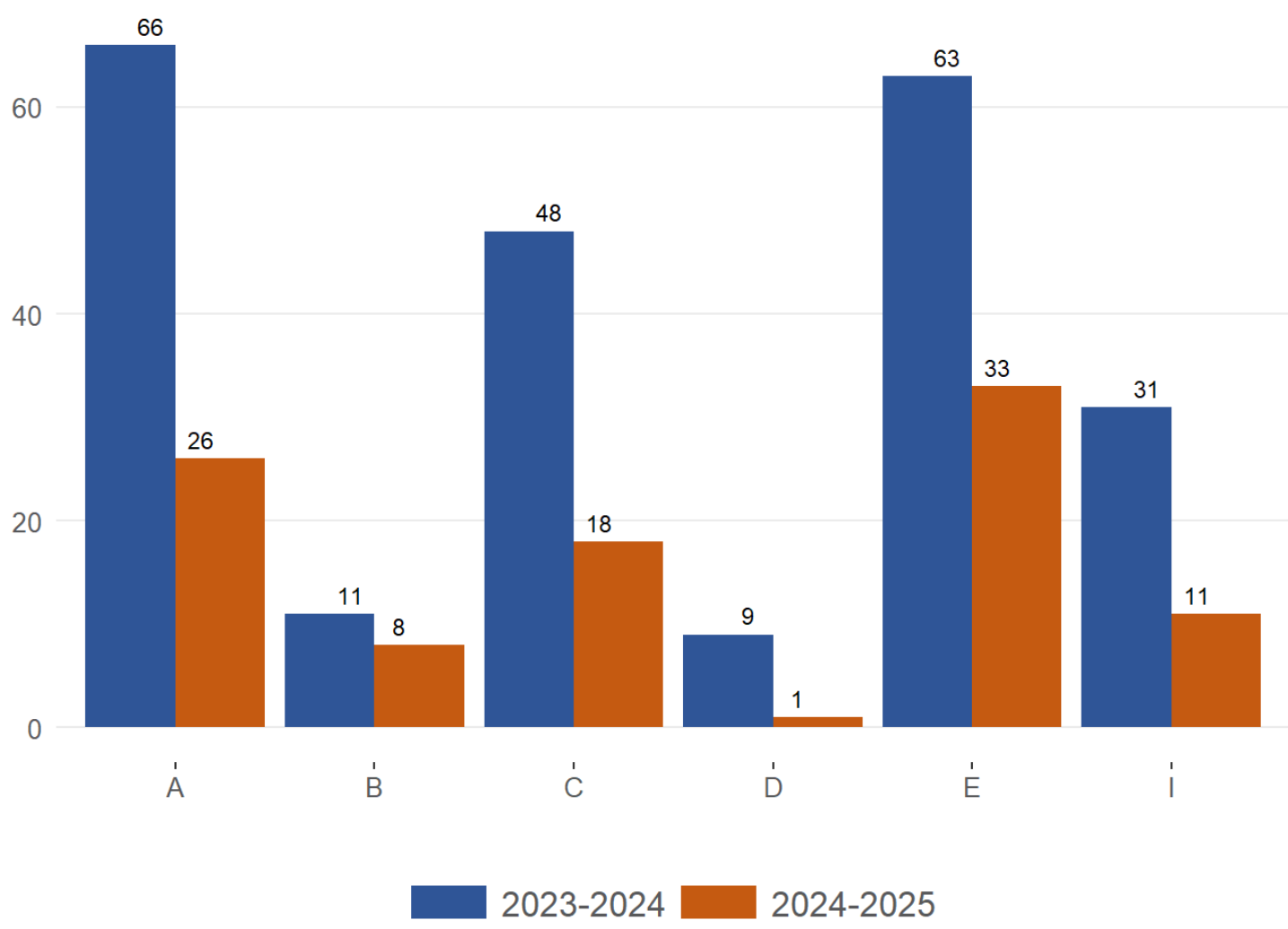
RESULT

Reduction of 41% of pediatric RSV hospitalizations (3342 avoided cases), 43% in the peak month (1796 cases avoided). Reduction mostly concerns eligible children <6 months-old



Cumulative incidences of pediatric (<5yo) hospitalizations for RSV per 1000 inhabitants, for both seasons (RSVPed)

Number of pediatric RSV hospitalizations per age group, for both seasons (RSVPed)



Number of pediatric (<5yo) admissions in ICU for RSV in 6 hospitals, for both seasons (RSVPed/SARI)

Reduction of 57% the burden of ICU admissions in 6 of 10 Belgian hospitals with a PICU

Effectiveness is >80%, in line with other studies² and doesn't present significant decrease up to 90 days after immunization

	Cases treated by nirsevimab	Control treated by nirsevimab	Cases not treated by nirsevimab	Cases not treated by nirsevimab	IE (adjusted) (95% CI)
Waning effect					
2-30 days	14	36	142	59	82.6% (60.1–92.4)
30-60 days	19	56	142	59	86.6% (73.6–93.2)
60-90 days	9	41	142	59	87.6% (69.9–94.9)
more than 90 days	2	27	142	59	89.5% (48.7–97.8)

Effectiveness of nirsevimab, estimated using logistic regression (test-negative design), depending on time since immunization (rows)(SARI)

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CONCLUSIONS

- Introduction of nirsevimab has already had a marked impact on the pediatric RSV burden in Belgium
- Data on prevention coverage and acceptability among parents are under investigation to guide upcoming preventive campaigns
- As several prevention methods will be available in the near future, namely a new maternal vaccine that is now reimbursed in Belgium and new monoclonal antibodies, this analysis will be repeated for the coming season(s)

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